August 19, 2025

[3]:
```python
import torch.nn as nn

layer = nn.Linear(10, 5)
```

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[5]:
```python
nn.init.zeros_(layer.weight)
nn.init.zeros_(layer.bias)
layer.weight, layer.bias
```

[5]:
```
(Parameter containing:
 tensor([[0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
         [0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
         [0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
         [0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
         [0., 0., 0., 0., 0., 0., 0., 0., 0., 0.]], requires_grad=True),
 Parameter containing:
 tensor([0., 0., 0., 0., 0.], requires_grad=True))
```

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[6]:
```python
#
nn.init.uniform_(layer.weight, -0.1, 0.1)
print(layer.weight)
#
nn.init.normal_(layer.weight, mean=0.0, std=0.01)
print(layer.weight)
```

```
Parameter containing:
tensor([[-0.0685, -0.0105, -0.0856,  0.0056, -0.0295,  0.0679, -0.0618, -0.0876,
         -0.0893,  0.0781],
        [ 0.0579, -0.0760,  0.0969, -0.0809, -0.0331, -0.0724, -0.0312,  0.0411,
          0.0185, -0.0560],
        [ 0.0598, -0.0355, -0.0625, -0.0768,  0.0403,  0.0676, -0.0613, -0.0126,
          0.0189, -0.0014],
        [-0.0258,  0.0902,  0.0432,  0.0546,  0.0029,  0.0046,  0.0578,  0.0752,
          0.0795,  0.0900],
        [-0.0214, -0.0078,  0.0949,  0.0321, -0.0423,  0.0825, -0.0948,  0.0605,
         -0.0232,  0.0135]], requires_grad=True)
```

```
Parameter containing:
tensor([[-0.0013,  0.0141, -0.0058, -0.0065, -0.0061, -0.0184,  0.0166,  0.0063,
         -0.0112, -0.0008],
        [-0.0146,  0.0023,  0.0102,  0.0024, -0.0002,  0.0118,  0.0083,  0.0048,
          0.0054, -0.0018],
        [-0.0052, -0.0160, -0.0138,  0.0078,  0.0097, -0.0110, -0.0261, -0.0136,
         -0.0021,  0.0041],
        [ 0.0104, -0.0256, -0.0125, -0.0082, -0.0087,  0.0021,  0.0150,  0.0096,
          0.0114, -0.0070],
        [ 0.0068,  0.0110,  0.0116,  0.0240,  0.0039,  0.0061, -0.0046, -0.0049,
         -0.0024, -0.0051]], requires_grad=True)
```

- Xavier

[7]:
```
#
nn.init.xavier_uniform_(layer.weight)
print(layer.weight)

#
nn.init.xavier_normal_(layer.weight)
print(layer.weight)
```

```
Parameter containing:
tensor([[-0.6109, -0.2448, -0.0749,  0.3422,  0.0735, -0.2993, -0.4203,  0.4667,
         -0.0193, -0.5302],
        [-0.5263, -0.2733, -0.4304, -0.4026, -0.1517,  0.3819,  0.4796,  0.4438,
         -0.3958, -0.2701],
        [ 0.2444, -0.4521, -0.0379,  0.3247, -0.2814, -0.5621, -0.1393,  0.4778,
         -0.4532, -0.0447],
        [ 0.0236,  0.4968,  0.4877,  0.2788,  0.5950,  0.1563, -0.2551, -0.4354,
         -0.4284,  0.0198],
        [ 0.2507, -0.3993,  0.5542,  0.4704, -0.1796, -0.2554, -0.0603,  0.3367,
         -0.6288,  0.3575]], requires_grad=True)
Parameter containing:
tensor([[ 0.5388, -0.0571, -0.3049, -0.2705, -0.1889,  0.1201,  0.1843,  0.9506,
         -0.2163, -0.2346],
        [-0.6500, -0.1329,  0.0328,  0.0947, -0.0942,  0.2541,  0.5549, -0.2671,
         -0.0390,  0.3599],
        [-0.2510, -0.1784, -0.0822, -0.7670, -0.3515, -0.4224,  0.2167,  0.2175,
         -0.3990,  0.1549],
        [-0.1890, -0.0818, -0.3432,  0.4186,  0.5504,  0.2774, -0.6545,  0.0431,
         -0.1896,  0.0967],
        [ 0.2659,  0.2529, -0.8084, -0.1266,  0.1223, -0.2236,  0.5855, -0.6044,
          0.1976, -0.0115]], requires_grad=True)
```

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[9]:
```
#
nn.init.kaiming_uniform_(layer.weight, nonlinearity='relu')
```

```
print(layer.weight)

#
nn.init.kaiming_normal_(layer.weight, nonlinearity='relu')
print(layer.weight)
```

```
Parameter containing:
tensor([[-0.0774, -0.2932, -0.5114,  0.6988, -0.5929, -0.2850,  0.2552, -0.6129,
          0.0163,  0.5273],
        [ 0.6227, -0.5041, -0.5457, -0.3972,  0.4128, -0.7044, -0.4219,  0.2629,
         -0.2613,  0.1805],
        [ 0.0012, -0.2870,  0.2125,  0.3865,  0.7463, -0.4740, -0.4688, -0.2381,
          0.5716,  0.3329],
        [-0.0358,  0.5264,  0.6922, -0.4462, -0.0600,  0.3052, -0.1537,  0.7180,
          0.1787,  0.0051],
        [-0.7085,  0.3538,  0.3950,  0.1785, -0.6982, -0.0302,  0.6342,  0.1716,
          0.7462, -0.7297]], requires_grad=True)
Parameter containing:
tensor([[ 0.1578, -1.1899,  0.4835,  0.1402, -0.5692,  0.2686,  0.3894,  0.1890,
         -0.7624, -0.7417],
        [ 0.2420,  0.5058,  0.4894,  0.5203, -0.1793, -0.2027,  0.2519, -0.1688,
         -0.6759, -0.6575],
        [-0.2185,  0.6618, -0.3759, -0.0283, -0.0336,  0.7114,  0.1026,  0.1616,
         -0.5968, -0.3482],
        [-0.5439, -0.9198, -0.0267,  0.3205,  0.2042,  0.1277,  0.0057,  0.3411,
         -1.0231,  0.0098],
        [ 0.0380,  0.6246, -0.3778,  0.4134,  0.1576,  0.0646, -0.8085,  0.0350,
          0.0196, -0.7902]], requires_grad=True)
```

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[10]:
```
nn.init.constant_(layer.bias, 0.0)
print(layer.bias)
```

```
Parameter containing:
tensor([0., 0., 0., 0., 0.], requires_grad=True)
```